higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## PROFESSIONAL ENGINEERING PRACTICE
### NQF LEVEL 4

(6021014)

**13 November 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 9 pages.**

Copyright reserved

**TIME: 3 HOURS**
**MARKS: 100**

---

**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

---

Copyright reserved 

## QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 How many atoms does an ozone molecule have?

A 1
B 5
C 2
D 3

1.1.2 In which layer of the atmosphere do we find the ozone layer?

A Mesosphere
B Thermosphere
C Troposphere
D Stratosphere

1.1.3 The function of the ozone layer is to …

A protect the earth from the harmful ultraviolet rays of the sun.
B harm humans and animals.
C enhance global warming.
D damage vegetation and crops.

1.1.4 Which of the following gases can deplete the ozone layer in the atmosphere?

A Carbon dioxide
B Nitrous oxide
C Sulphur dioxide
D Carbon monoxide

1.1.5 The chemical symbol for ozone:

A $O_2$
B $CO_2$
C $O_3$
D CO

(5 × 1) (5)

1.2 Chemical waste must be properly disposed of, as it can be extremely hazardous to humans, animals and elements such as metals. Chemical waste is hazardous due to the toxic or poisonous properties thereof.

Name FOUR other properties of chemical waste that makes it hazardous. (4)

1.3 You are the manager in charge of a workshop.

Apart from salary increases and raises, state SEVEN other factors that you will implement to ensure that the workers experience job satisfaction. (7)

1.4 In South Africa, many heavy vehicles on the main roads cause traffic congestion.

State FOUR negative consequences that the congestion caused by many heavy vehicles on our main roads have on other road users and the environment. (4)

1.5 The Engineering Council of South Africa (ECSA) regulates the Engineering profession in South Africa.

State FIVE objectives of ECSA. (5)

**[25]**

## QUESTION 2

2.1 Discuss the following steps of the engineering design process:

2.1.1 Generation of concept solutions through brainstorming

2.1.2 Detail design of the selected concept solution

2.1.3 Building and testing the prototype

(3 × 4) (12)

2.2 A concept selection matrix is used to select the best possible solution to a design problem from a small number of possible concept solutions.

Study and use the concept selection matrix below to answer the questions.

| CRITERIA | WEIGHTING | CONCEPT 1 | | CONCEPT 2 | | CONCEPT 3 | |
|---|---|---|---|---|---|---|---|
| Cost | 5 | 6 | 30 | 2 | 10 | 9 | 45 |
| Safety | (2.2.1) | 6 | 54 | 5 | 46 | 2 | 18 |
| Ergonomics | 3 | 6 | 18 | (2.2.2) | 21 | 2 | 6 |
| Reliability | 7 | 8 | (2.2.3) | 1 | 7 | 7 | 49 |
| Total score | – | 158 | | 83 | | (2.2.4) | |

2.2.1 What is the weighting factor for safety? (1)

2.2.2 What is the rating factor of concept 2 for ergonomics? (1)

2.2.3 What is the criterion total for concept 1 for reliability? (1)

2.2.4 What is the final concept score for concept 3? (1)

Copyright reserved 

2.2.5 Which of the concepts is regarded as the most reliable? (1)

2.2.6 Supply a reason for your answer to QUESTION 2.2.5. (The answer must be deduced from the matrix above.) (1)

2.2.7 The matrix contains one error. Indicate the error and explain how it should be corrected. (2)

2.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.3.1–2.3.5) in the ANSWER BOOK.

2.3.1 Mathematical modelling uses equations to describe a physical reality.

2.3.2 A mock-up is an example of a mathematical model.

2.3.3 Prototypes are used to demonstrate the functioning of a product but cannot be used for testing purposes.

2.3.4 A simulation is a computer program that mimics a real process.

2.3.5 Scale models are often used in wind-tunnel testing.

(5 × 1) (5)
**[25]**

**QUESTION 3**

3.1 Complete the following sentences by using the words provided in the list below. Write only the word(s) next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

| |
|---|
| freehand; symmetry; thin; cutting plane; smaller; thick; isometric; tolerance; perspective; larger; dimension |

3.1.1 In … projection the edges are drawn at an angle of 30° to the horizontal.

3.1.2 A dimension line used in drawings is a continuous … line.

3.1.3 When irregular boundaries are indicated on a drawing, a continuous, thin … line is used.

3.1.4 When a scale of 1:3 is used in a drawing, the drawing on the drawing sheet will be … than the real size of the component.

3.1.5 The permissible amount of deviation allowed in the size of a component is called it's …

(5 × 1) (5)

3.2 A technical report consists of various sections, such as the title page, introduction, et cetera. Discuss the information that the intended reader of the report should expect under the following section headings in detail:

3.2.1 Introduction 

3.2.2 Conclusion

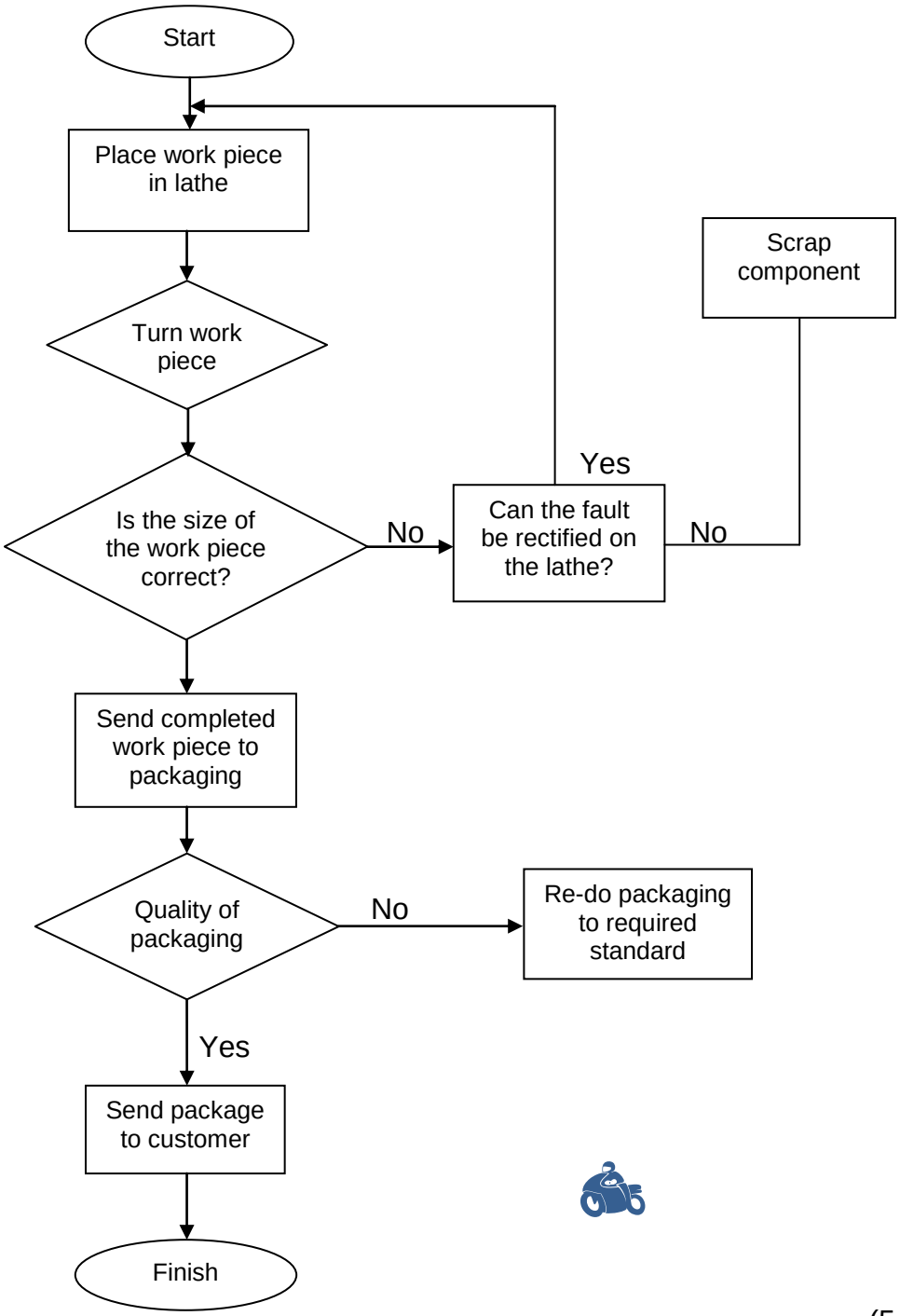
3.2.3 Recommendations

3.2.4 Appendices 

(4 × 2) (8)

3.3 The flow chart below describes the flow process of a specific component from the lathe to packaging for the customer. Five errors appear in the flow chart.

Identify the FIVE errors by analysing the flow chart. Also indicate how each error should be corrected.




(5 × 2) (10)

3.4 A flow chart is ONE method of design system representation.

Name TWO other methods of design system representation. (2)

**[25]**

## QUESTION 4

4.1 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–M) next to the question number (4.1.1–4.1.10) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 4.1.1 | Combination of statements and variables that need to be evaluated, such as 'a = b + c' | A | +, -, >, <, etc. |
| 4.1.2 | Signs used to perform operations | B | variables |
| 4.1.3 | Transfers information to computer | C | input device |
| 4.1.4 | Counting in twos instead of tens | D | bit |
| 4.1.5 | Receives processed data from the computer | E | low-level language |
| 4.1.6 | Hard drive found in computer is an example | F | high-level language |
| 4.1.7 | Regulates flow of data between ALU and RAM | G | IF statement |
| 4.1.8 | Letters used to hold specific information | H | case statement |
| 4.1.9 | Understands machine code only | I | storage device |
| 4.1.10 | Two states of a transistor – On or Off | J | expressions |
| | | K | binary format |
| | | L | output device |
| | | M | control unit |

(10 × 1) (10)

4.2 The Waterfall Model is ONE example of complex software systems.

Name TWO other examples of complex software systems. (2)

4.3 The ALU of a computer can perform bitwise logic operations such as the XOR operation.

Name THREE other bitwise logic operations that the ALU can perform. (3)

Copyright reserved 

4.4 State whether the following devices are input, output or storage devices. Only write 'Input', 'Output' or 'Storage' next to the question number (4.4.1–4.4.4) in the ANSWER BOOK.

4.4.1 Scanner

4.4.2 Smart card

4.4.3 Blu-ray disc

4.4.4 Plotter

(4 × 1) (4)

4.5 The spreadsheet below indicates the enrolment figures at a TVET college in South Africa for the NC(V) programme since 2015. Study the information and answer the questions.

| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 2 | | ABC TVET COLLEGE – ENROLMENT FIGURES (ERD) | | | | | | |
| 3 | | | | | | | | |
| 4 | ***Year/Programme*** | ***2015*** | | ***2016*** | | ***2017*** | | |
| 5 | | | | | | | | |
| 6 | Fitting and Turning | 80 | | 65 | | 44 | | 189 |
| 7 | Welding | 76 | | 80 | | 79 | | 235 |
| 8 | Boilermaking | 89 | | 90 | | 95 | | 274 |
| 9 | ARM | 35 | | 37 | | 34 | | 106 |
| 10 | | | | | | | | |
| 11 | Total | 280 | | 272 | | 252 | | |
| 12 | Average | 75 | | 73 | | 63 | | |

4.5.1 Which TWO font styles are used in cell B4? (2)

4.5.2 In which programme was a strong decline in numbers visible for the period indicated in the spreadsheet? (1)

4.5.3 Give a reason for your answer to QUESTION 4.5.2. (1)

4.5.4 Write down the range formula used to calculate the data displayed in cell F12. (2)

**[25]**

**TOTAL: 100**

Copyright reserved